Learn Practice Improve Grow

Python Dictionaries

Practice Question

Small Steps Big Skills!

Keep Practicing Python!

3 Question 3

What will be the output of the following code?

```
info = {"name": "Aman", "age": 19, "course": "BPT"}
info["age"] = 20
info["city"] = "Patna"
del info["course"]
print(info)
```

A `{'name': 'Aman', 'age': 20, 'city': 'Patna'}`

B `{'name': 'Aman', 'age': 19, 'city': 'Patna'}`

C `{'name': 'Aman', 'age': 20, 'course': 'BPT', 'city': 'Patna'}`

D `{'age': 20, 'city': 'Patna'}`

Key Concept:

- You can add a new key-value pair using `dict[key] = value`.
- You can update the value of an existing key.
- You can remove a key using `del dict[key]`.
- The dictionary will only contain the remaining key-value pairs.

Practice Build Concepts Do Better!

"A Small Step in Practice, A Big Step Towards Success!"

By Satyarthi

- [ ] Think
- [ ] Write Code
- [ ] Test
- [ ] Improve
- [ ] Keep Practicing

Python Functions

Practice Coding Questions

Practice Today → Build a Better Tomorrow

Simple
Powerful
Versatile

By Satyarthi

1 Write a function that takes a string as input and returns the number of vowels in the string.

Example :

- Input : "python"
- Output : 1
  (Only 'o' is a vowel)
- Input : "Hello World"
- Output : 3
  (e, o, o)

Vowels are: a, e, i, o, u (consider both lowercase and uppercase).

Code :

```
1  def count_vowels(s):
2      # write your code here
3
4
5  # Example usage
6  text = "Hello World"
7  print(count_vowels(text))
8
```

Expected Output :

```
3
```

Write Functions, Write a Better Future

2 Write a function that takes a list of numbers and returns the largest number from the list.

Example :

- Input : [3, 7, 2, 9, 5]
- Output : 9
- Input : [10, -1, 6, 4]
- Output : 10

Try to solve it without using the built-in function max().

Code :

```
1  def find_largest(nums):
2      # write your code here
3
4
5  # Example usage
6  numbers = [3, 7, 2, 9, 5]
7  print(find_largest(numbers))
8
```

Expected Output :

```
9
```

"Functions make your code reusable, organized and powerful." – Satyarthi

- [ ] Learn
- [ ] Practice
- [ ] Solve
- [ ] Improve
- [ ] Be Confident

Python Functions

Practice Coding Questions

Small Functions → Big Learning

Simple
Powerful
Versatile

By Satyarthi

3 Write a function that takes a number as input and returns its factorial.

Example :

- Input : 5
- Output : 120
  (5! = 5 × 4 × 3 × 2 × 1 = 120)
- Input : 0
- Output : 1
  (0! is defined as 1)

Factorial of a number n is the product of all positive integers from 1 to n.

Code :

```
1  def factorial(n):
2      # write your code here
3
4
5  # Example usage
6  print(factorial(5))
7
```

Expected Output :

```
120
```

Break Problems into Functions

4 Write a function that takes a string as input and returns the string in reverse order.

Example :

- Input : "python"
- Output : "nohtyp"
- Input : "Hello World"
- Output : "dlroW olleH"

Strings in Python can be reversed using slicing or a loop.

Code :

```
1  def reverse_string(s):
2      # write your code here
3
4
5  # Example usage
6  print(reverse_string("python"))
```

Expected Output :

```
nohtyp
```

Practice More Code Better

"A function is a block of code that makes your program smarter." – Satyarthi

- [ ] Learn
- [ ] Practice
- [ ] Solve
- [ ] Improve
- [ ] Grow

Python Functions

Practice Coding Questions

Practice Functions → Stronger Python Skills

Simple
Powerful
Versatile

By Satyarthi

5 Write a function that takes a list of numbers as input and returns the sum of only the positive numbers.

Example :

- Input : [1, -2, 3, -4, 5]
- Output : 9
  (1 + 3 + 5 = 9)
- Input : [-1, -3, -5]
- Output : 0
  (No positive numbers)

Use a loop or list comprehension to add only numbers greater than 0.

Code :

```
1  def sum_positive(nums):
2      # write your code here
3
4
5  # Example usage
6  print(sum_positive([1, -2, 3, -4, 5]))
7
```

Expected Output :

```
9
```

Focus on Logic Not Just Code

6 Write a function that takes a string as input and returns the number of words in the string.

Example :

- Input : "Hello World"
- Output : 2
- Input : "Python is fun"
- Output : 3
- Input : " Practice makes perfect "
- Output : 3

You can use the split() function to separate words in a string.

Code :

```
1  def count_words(s):
2      # write your code here
3
4
5  # Example usage
6  print(count_words("Hello World"))
7
```

Expected Output :

```
2
```

"Practice writing functions — it makes your code cleaner and your thinking stronger." — Satyarthi

- [ ] Plan
- [ ] Write Code
- [ ] Test
- [ ] Debug
- [ ] Improve

Python Functions

Practice Coding Questions

Practice Functions → Build Problem Solving Skills

Simple
Powerful
Versatile

By Satyarthi

7 Write a function that takes a number as input and returns True if it is a prime number, otherwise False.

Example :

- Input : 7
- Output : True
- Input : 10
- Output : False
- Input : 2
- Output : True

A prime number is a number greater than 1 that has exactly two divisors: 1 and itself.

Code :

```
1  def is_prime(n):
2      # write your code here
3      pass
4
5  # Example usage
6  print(is_prime(7))
```

Expected Output :

```
True
```

Think
Logically
Write Clean Code

8 Write a function that takes a list of numbers as input and returns the average of the numbers.

Example :

- Input : [10, 20, 30]
- Output : 20.0
- Input : [5, 15]
- Output : 10.0
- Input : [1, 2, 3, 4]
- Output : 2.5

$$\text{Average} = \frac{\text{(Sum of numbers)}}{\text{(Total count)}}$$

Code :

```
1  def find_average(nums):
2      # write your code here
3
4
5  # Example usage
6  print(find_average([10, 20, 30]))
```

Expected Output :

```
20.0
```

Small
Functions
Big Results

"The best way to learn programming is by writing functions and solving real problems." – Satyarthi

- [ ] Learn
- [ ] Practice
- [ ] Solve Problems
- [ ] Build Logic
- [ ] Be Consistent

Python Functions

Practice Coding Questions

Small Steps → Big Programming Skills

Simple
Powerful
Versatile

By Satyarthi

11 Write a function that takes a list of numbers as input and returns the largest number.

Example :

- Input : [4,8, 1, 9, 3]
- Output : 9
- Input : [-5, -1, -8, -3]
- Output : -1
- Input : [10]
- Output : 10

Try to solve it without using the built-in function max().

Code :

```
1  def find_largest(nums):
2      # write your code here
3
4
5  # Example usage
6  numbers = [4, 8, 1, 9, 3]
7  print(find_largest(numbers))
8
```

Expected Output :

```
9
```

Solve Step by Step and Strengthen Your Logic

12 Write a function that takes a string as input and returns True if it is a palindrome, otherwise False.

Example :

- Input : "madam"
- Output : True
- Input : "racecar"
- Output : True
- Input : "python"
- Output : False

A palindrome is a word, number, or string that reads the same forward and backward.

Code :

```
1  def is_palindrome(s):
2      # write your code here
3
4
5  # Example usage
6  print(is_palindrome("madam"))
7  print(is_palindrome("python"))
```

Expected Output :

```
True
False
```

Practice Functions Build Better Programs

"Functions help you write once, use many times." — Satyarthi

- [ ] Learn
- [ ] Practice
- [ ] Solve Problems
- [ ] Build Skills
- [ ] Grow

Python Functions

Practice Coding Questions

Simple
Powerful
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By Satyarthi

Practice More → Build Stronger Logic → Become Better

15 Write a function that takes a list of numbers as input and returns a new list with the squares of each number.

Example :

- Input : [1, 2, 3, 4, 5]
- Output : [1, 4, 9, 16, 25]
- Input : [-2, 0, 3]
- Output : [4, 0, 9]
- Input : []
- Output : []

Use a loop (or list comprehension) to find the square of each number (n ** 2).

Code :

```
1  def get_squares(nums):
2      # write your code here
3
4
5  # Example usage
6  print(get_squares([1, 2, 3, 4, 5]))
```

Expected Output :

```
[1, 4, 9, 16, 25]
```

Turn Simple Logic into Useful Functions

16 Write a function that takes a string as input and returns the number of vowels (a, e, i, o, u) in the string (cese-insensitive).

Example :

- Input : "Hello"
- Output : 2 (e, o)
- Input : "PYTHON"
- Output : 1 (o)
- Input : "aeiou"
- Output : 5
- Input : "bcdfg"
- Output : 0

Convert the string to lowercase and check each character in a loop.

Code :

```
1  def count_vowels(s):
2      # write your code here
3
4
5  # Example usage
6  print(count_vowels("Hello"))
7  print(count_vowels("PYTHON"))
```

Expected Output :

```
2
1
```

Functions Make Repetitive Tasks Easy

"Practice functions today, create better programs tomorrow." — Satyarthi

- [ ] Learn
- [ ] Practice
- [ ] Solve
- [ ] Build Logic
- [ ] Be a Better Coder

Python
Nested Functions
Practice Coding Questions

Functions Inside Functions → More Powerful Programs

Simple
Powerful
Versatile

By Satyarthi

What are Nested Functions?

A function defined inside another function is called a nested function. The inner function can access variables of the outer function.

Why use Nested Functions?

- ✅ Helps in code organization
- ✅ Keeps helper functions private
- ✅ Useful in closures and decorators
- ✅ Makes code more readable

Small Concepts
Big Possibilities

1 Write a program using nested functions to calculate the area and perimeter of a rectangle.

Problem Statement:

Define a function rectangle(a, b) that takes length and breadth as input. Inside it, define two nested functions:

- One to calculate the area (length × breadth)
- Another to calculate the perimeter (2 × (length + breadth))

The outer function should return both the area and perimeter as a tuple.

Code :

```
1  def rectangle(a, b):
2      def area():
3          return a * b
4
5      def perimeter():
6          return 2 * (a + b)
7
8      return area(), perimeter()
9
10 # Example usage
   print(rectangle(5, 3))
```

Expected Output :

```
(15, 16)
```

Explanation :

- area() and perimeter() are nested functions.
- They can access a and b from the outer function.
- The outer function returns a tuple containing both values.

Try changing the input values and check the result.

2 Write a program using nested functions to check whether a number is even, odd, or zero.

Problem Statement:

Define a function check_number(n) that takes a number as input. Inside it, define three nested functions:

- One to check if the number is zero
- One to check if the number is even
- One to check if the number is odd

The outer function should call these nested functions and return an appropriate message.

Code :

```
1  def check_number(n):
2      def is_zero():
3          return n == 0
4
5      def is_even():
6          return n % 2 == 0
7
8      def is_odd():
9          return n % 2 != 0
10
11     if is_zero():
12         return "Number is Zero"
13     elif is_even():
14         return "Number is Even"
15     else:
           return "Number is Odd"

   # Example usage
   print(check_number(7))
```

Example Outputs:

Input: 0 → Output: Number is Zero
Input: 4 → Output: Number is Even
Input: 7 → Output: Number is Odd

Explamation:

- is_zero(), is_even(), and is_odd() are nested functions.
- They check the condition using the outer function's variable n.
- The outer function returns the appropriate message.

Try different input values (positive, negative, zero) to test your program.

"Functions within functions create powerful solutions." – Satyarthi

- [ ] Learn
- [ ] Practice
- [ ] Solve Problems
- [ ] Build Logic
- [ ] Be Consistent

Python
Nested Functions
Practice Coding Questions

Small Functions → Big Solutions

Simple
Powerful
Versatile

By Satyarthi

5 Write a program using nested functions to check whether a given string is a palindrome.

Problem Statement:

Define a function is_palindrome(s) that takes a string as input. Inside it, define a nested function clean_text(text) which removes spaces and converts the string to lowercase.

The outer function should use this nested function to check whether the cleaned string is a palindrome or not.

Return True if it is a palindrome, otherwise False.

Nested functions can help in breaking a big task into smaller, reusable steps.

Code :

```
1  def is_palindrome(s):
2      def clean_text(text):
3          return text.replace(" ", "").lower()
4
4      def check_palindrome(text):
5          return text == text[::-1]
7
8      cleaned = clean_text(s)
8      return check_palindrome(cleaned)
9
10
11 # Example usage
   print(is_palindrome("A man a plan a canal Panama"))
   print(is_palindrome("Hello World"))
```

Expected Output :

```
True
False
```

Explanation:

- clean_text(text) removes spaces and converts the string to lowercase.
- check_palindrome(text) checks if the string is equal to its reverse.
- The outer function uses both nested functions to return the result.

Try with different inputs like "Madam", "Race Car", "Python" etc.

6 Write a program using nested functions to count the number of uppercase, lowercase, digits and special characters in a string.

Problem Statement:

Define a function analyze_string(s) that takes a string as input. Inside it, define four nested functions:

1. count_upper(text)
2. count_lower(text)
3. count_digits(text)
4. count_special(text)

The outer function should call these nested functions and return a dictionary with the count of each type of character.

Nested functions help in organizing related tasks inside one function.

Code :

```
1  def analyze_string(s):
2      def count_upper(text):
3          return sum(1 for ch in text if ch.isupper())
4
5      def count_lower(text):
5          return sum(1 for ch in text if ch.islower())
6
7      def count_digits(text):
7          return sum(1 for ch in text if ch.isdigit())
9
10     def count_special(text):
11         return sum(1 for ch in text if not ch.isalnum())
12
13     return {
14         "Uppercase": count_upper(s),
15         "Lowercase": count_lower(s),
15         "Digits": count_digits(s),
16         "Special Characters": count_special(s)
       }

   # Example usage
   print(analyze_string("Hello@123"))
```

Expected Output :

```
{'Uppercase': 1,
 'Lowercase': 4,
 'Digits': 3,
 'Special Characters': 1}
```

Explanation:

- Each nested function counts a specific type of character.
- The outer function combines all the counts in a dictionary and returns it.
- This shows how nested functions can be used to keep related logic together.

Try with different strings like "PYTHON_2026!" or "Hello World".

"Good code is written with small, well-organized steps." — Satyarthi

- [ ] Learn
- [ ] Practice
- [ ] Solve Problems
- [ ] Build Logic
- [ ] Be a Better Coder

Python
Nested Functions
Practice Coding Questions

Practice Today → Stronger Programming Tomorrow

Simple
Powerful
Versatile

By Satyarthi

7 Write a program using nested functions to calculate the power of a number (n^p).

Problem Statement:

Define a function power(n, p) that takes a number n and a non-negative integer p as input. Inside it, define a nested function multiply(a, b) which multiplies two numbers.

Use the nested function to calculate n raised to the power p (without using the ** operator).

Return the result.

This helps you understand how nested functions can be used for repeated operations.

Code :

```python
def power(n, p):
    def multiply(a, b):
        return a * b

    result = 1
    for i in range(p):
        result = multiply(result, n)
    return result

# Example usage
print(power(2, 5))
```

Expected Output :

```
32
```
(for input 2, 5)

Explanation :

- The nested function multiply(a, b) multiplies two numbers.
- It is used repeatedly to calculate n^p.
- The outer function returns the final result.

Try with different inputs like (3, 4) or (5, 3).

8 Write a program using nested functions to find the largest number in a list.

Problem Statement:

Define a function find_max(numbers) that takes a list of numbers as input. Inside it, define a nested function is_greater(a, b) which returns True if a is greater than b, otherwise False.

Use the nested function to find and return the largest number from the list.

Nested functions can be used to break a problem into smaller logical checks.

Code :

```python
def find_max(numbers):
    def is_greater(a, b):
        return a > b

    max_num = numbers[0]
    for num in numbers[1:]:
        if is_greater(num, max_num):
            max_num = num
    return max_num

# Example usage
print(find_max([3, 7, 2, 9, 5]))
```

Expected Output :

```
9
```
(for input [3, 7, 2, 9, 5])

Explanation :

- The nested function is_greater(a, b) checks which number is larger.
- The outer function uses it to iterate through the list and keep track of the largest number.
- Finally, it returns the largest number.

Try with negative numbers or a list with one element.

"Nested functions make your code modular and easy to understand." — Satyarthi

- [ ] Learn
- [ ] Practice
- [ ] Solve Problems
- [ ] Build Logic
- [ ] Be a Better Coder

Python
Lambda Functions
Concepts Made Simple

Small Functions → Quick Use → Powerful Results

Simple
Powerful
Versatile

By Satyarthi

What is a Lambda Function?

A lambda function is a small **anonymous** (no name) function in Python. It is used for short, simple operations where a full function definition is not needed.

Syntax

```
lambda arguments: expression
```

- No function name
- Single expression (one line)
- Automatically returns the result

Key Points

- Used for short and simple tasks
- Can take any number of arguments
- Can be used with functions like map(), filter(), sorted()
- Makes code shorter and cleaner
- Cannot have multiple statements

1 Example 1: Square a Number

Using a lambda function to find the square of a number.

```
square = lambda x: x ** 2
print(square(5))
```

Output :
25

2 Example 2: Add Two Numbers

Using a lambda function to add two numbers.

```
add = lambda a, b: a + b
print(add(3, 7))
```

Output :
10

3 Example 3: Use with map()

Using lambda with map() to square each number in a list.

```
nums = [1, 2, 3, 4]
result = list(map(lambda x: x**2, nums))
print(result)
```

Output :
[1, 4, 9, 16]

4 Example 4: Use with filter()

Using lambda with filter() to get even numbers from a list.

```
nums = [1, 2, 3, 4, 5, 6]
even_nums = list(filter(lambda x: x % 2 == 0, nums))
print(even_nums)
```

Output :
[2, 4, 6]

5 Example 5: Use with sorted()

Using lambda to sort a list of tuples based on the second element.

```
data = [(1, 'A'), (3, 'C'), (2, 'B')]
sorted_data = sorted(data, key=lambda x: x[1])
print(sorted_data)
```

Output :
[(1, 'A'), (2, 'B'), (3, 'C')]

When to Use?

- When the function is small and used only once
- With higher-order functions like map(), filter(), sorted()
- To make code shorter and more readable

Remember :

"Use lambda functions for simple tasks, and regular functions for complex logic."

"Short functions can do big things." – Satyarthi